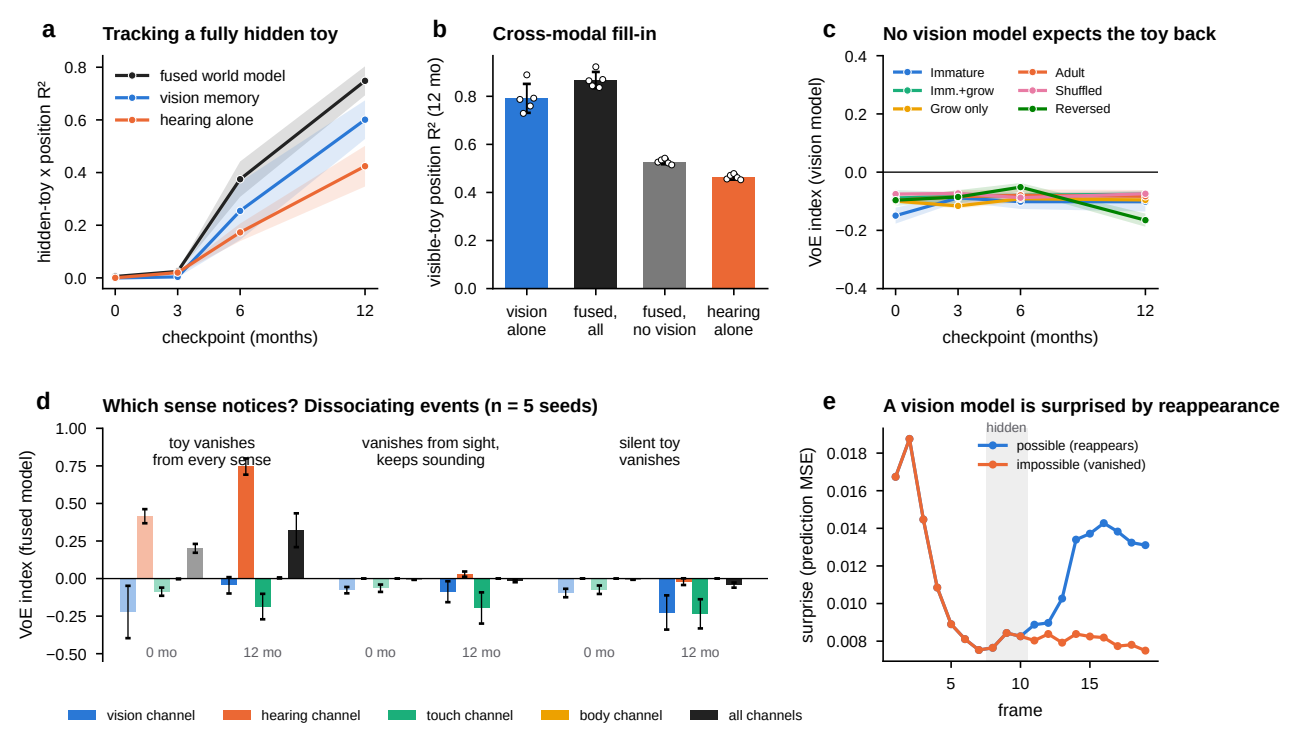
a
Tracking a fully hidden toy
fused world model
vision memory
hearing alone
hidden-toy x position R²
checkpoint (months)
b
Cross-modal fill-in
visible-toy position R² (12 mo)
vision alone
fused, all
fused, no vision
hearing alone
c
No vision model expects the toy back
Immature
Imm.+grow
Grow only
Adult
Shuffled
Reversed
VoE index (vision model)
checkpoint (months)
d
Which sense notices? Dissociating events (n = 5 seeds)
toy vanishes from every sense
vanishes from sight, keeps sounding
silent toy vanishes
VoE index (fused model)
0 mo
12 mo
vision channel
hearing channel
touch channel
body channel
all channels
e
A vision model is surprised by reappearance
hidden
possible (reappears)
impossible (vanished)
surprise (prediction MSE)
frame